Math Kangaroo Practice

Grade 3-4 · 20 questions · standard · seed 1019

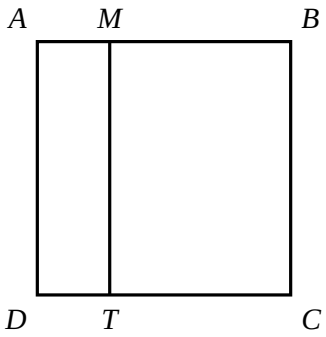
Part A: Each question is worth 3 points

1. Ben, Lucas, Aisha and Julien were born in different months: October, April, December and July. Aisha was born in autumn. Lucas was born in spring. Ben was born in a month that starts with J. Who was born in December?

A. Lucas ***B.*** Ben ***C.*** Julien

D. Aisha ***E.*** Impossible to determine

2. *ABCD* is a square. Its side is 7 cm. *AMTD* is a rectangle. Its shorter side is 2 cm.

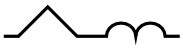
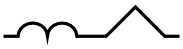


How many centimetres is the perimeter of the square *ABCD* larger than that of the rectangle *AMTD*?

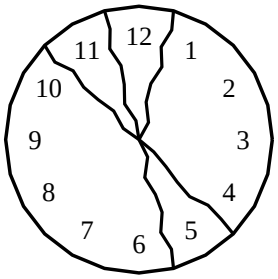
A. 10 cm ***B.*** 14 cm ***C.*** 18 cm ***D.*** 19 cm ***E.*** 20 cm

3. Lena cut a strip of paper along this line.

Which piece **cannot** come from the cut line?

A. ***B.***  ***C.*** ***D.***  ***E.***

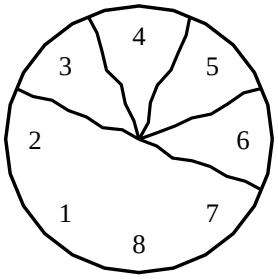
4. The face of a clock is cracked into 5 pieces as shown.



Which piece has the largest sum of numbers?

A. the piece with the 10 ***B.*** the piece with the 5 ***C.*** the piece with the 12
D. the piece with the 4 ***E.*** the piece with the 11

5. A round cake with the numbers 1 to 8 around its edge is cracked into 5 pieces as shown.

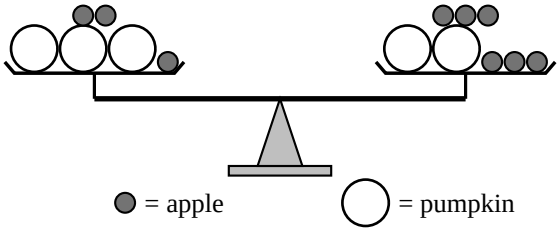


Which piece has the largest sum of numbers?

A. the piece with the 4 ***B.*** the piece with the 6 ***C.*** the piece with the 8
D. the piece with the 5 ***E.*** the piece with the 3

Part B: Each question is worth 4 points

6. The balance is level: 3 pumpkins and 3 apples on one plate, 2 pumpkins and 6 apples on the other.



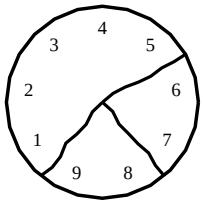
One pumpkin weighs the same as how many apples?

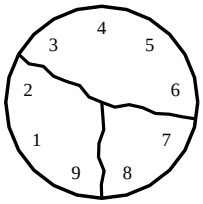
A. 2 apples ***B.*** 3 apples ***C.*** 4 apples ***D.*** 6 apples ***E.*** 9 apples

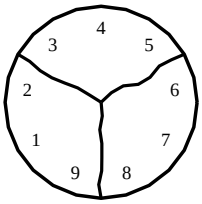
7. Chloe, Manon, Fabienne and Sofia were born in different months: March, October, June and July. Fabienne was not born in July. Manon was born in autumn. Chloe was born in spring.
Which statement is true?

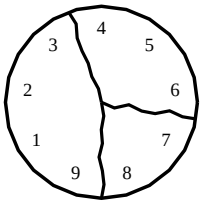
A. Chloe was born in October ***B.*** Sofia was born in March ***C.*** Manon was born in July
D. Fabienne was born in June ***E.*** Chloe was born in July

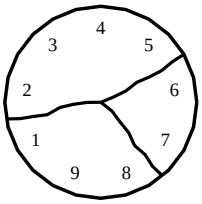
8. A round cake with the numbers 1 to 9 around its edge is cracked into 3 pieces. All the pieces have the same sum of numbers.
Which picture shows how it is cracked?

A. 

B. 

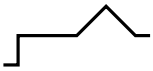
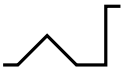
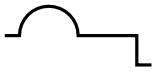
C. 

D. 

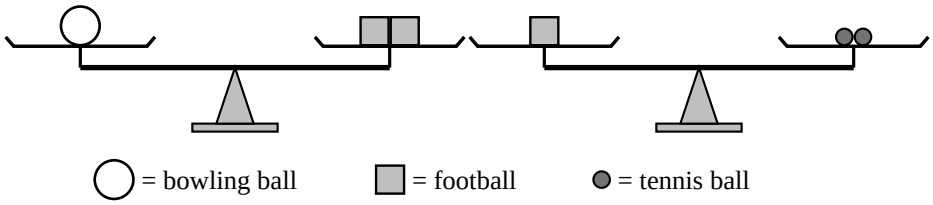
E. 

9. Far away we see the skyline of a castle.

Which of the pieces **can** belong to the skyline?

A.  ***B.*** ***C.***  ***D.*** ***E.*** 

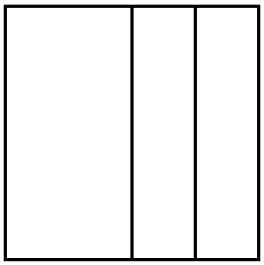
10. The two balances are level. One bowling ball weighs as much as 2 footballs, and one football weighs as much as 2 tennis balls.



How many tennis balls weigh as much as 1 bowling ball?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

11. Grandma bakes a square cake. A ribbon 36 cm long fits exactly around it. She cuts the cake into 3 rectangular pieces as shown and puts a ribbon around every piece.



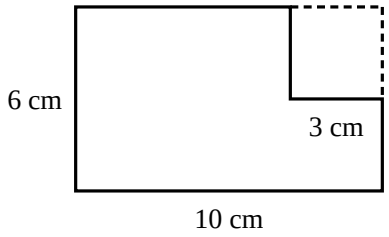
How many centimetres of ribbon does she need for all the pieces together?

A. 63 cm ***B.*** 72 cm ***C.*** 74 cm ***D.*** 77 cm ***E.*** 108 cm

12. Aisha, Hana, Emma and Julien each play a different instrument: the harp, the guitar, the flute and the drum. Julien blows into the instrument. Hana doesn't play the guitar. Aisha plays an instrument with strings. Hana plays an instrument with strings.
Which statement is true?

A. Aisha plays the harp ***B.*** Emma plays the flute ***C.*** Emma plays the harp
D. Aisha plays the guitar ***E.*** Aisha plays the flute

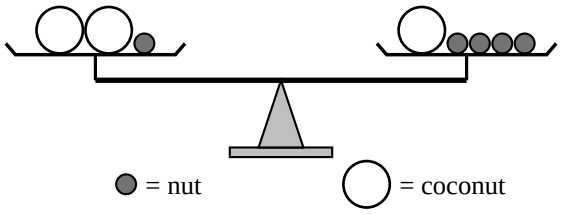
13. A rectangle is 10 cm long and 6 cm wide. A square with side 3 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 32 cm ***B.*** 34 cm ***C.*** 38 cm ***D.*** 39 cm ***E.*** 44 cm

14. The balance is level: 2 coconuts and 1 nut on one plate, 1 coconut and 4 nuts on the other.

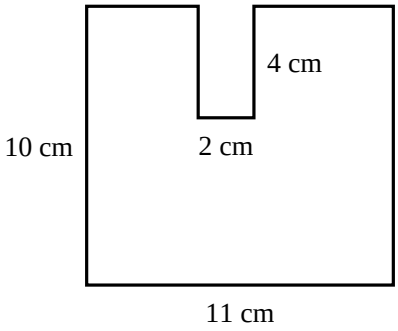


One coconut weighs the same as how many nuts?

A. 1 nut ***B.*** 2 nuts ***C.*** 3 nuts ***D.*** 4 nuts ***E.*** 5 nuts

Part C: Each question is worth 5 points

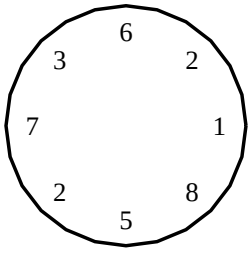
15. A 11 cm by 10 cm rectangle of cardboard has a rectangular notch 2 cm wide and 4 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 55 cm ***B.*** 54 cm ***C.*** 52 cm ***D.*** 50 cm ***E.*** 42 cm

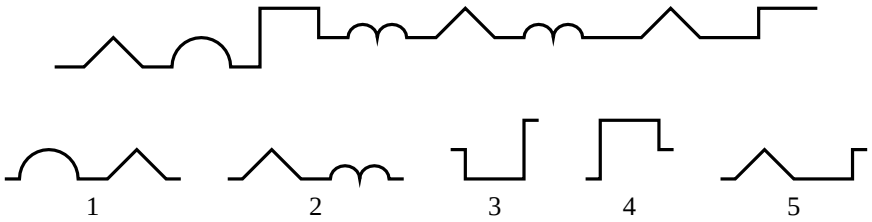
16. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

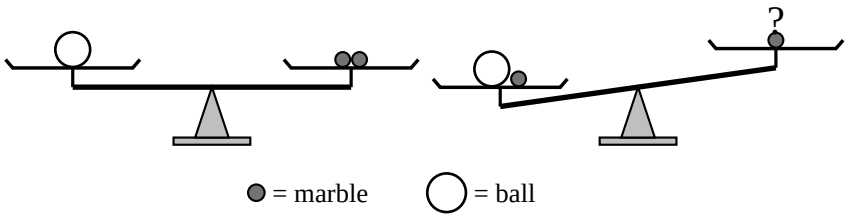
17. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

18. The first balance is level. On the second balance there is 1 ball and 1 marble on the left and 1 marble on the right.



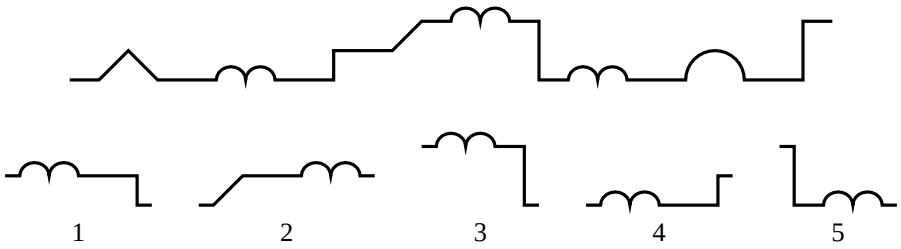
How many marbles must be added to the right plate to make it level?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

19. Emma, Leo, Tom and Fabienne each play a different instrument: the harp, the xylophone, the piano and the violin. Emma presses keys. Tom uses a bow. Leo never uses sticks.
Which statement is **not** true?

A. Leo plays the harp ***B.*** Fabienne plays the xylophone ***C.*** Emma plays the piano
D. Tom plays the violin ***E.*** Emma plays the violin

20. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

Answer Key

1.	**C**	**2.**	**A**	**3.**	**E**	**4.**	**A**	**5.**	**C**	**6.**	**B**
7.	**D**	**8.**	**D**	**9.**	**B**	**10.**	**B**	**11.**	**B**	**12.**	**D**
13.	**A**	**14.**	**C**	**15.**	**D**	**16.**	**B**	**17.**	**C**	**18.**	**A**
19.	**E**	**20.**	**C**								

Solutions

1. C (Julien) — The only possible matching: Ben – July; Lucas – April; Aisha – October; Julien – December. [2002 Q18 · who_has]

2. A (10 cm) — Square: 4 × 7 = 28 cm. Rectangle: 2 + 7 + 2 + 7 = 18 cm. Difference 28 – 18 = 10 cm. [2002 Q6 · square_minus_strip]

3. E — All the other pieces can be found in the line, but this one cannot: it is a mirror image of a part of the line. [2002 Q7 · cannot_belong]

4. A (the piece with the 10) — The sums are: 12 = 12, 1+2+3+4 = 10, 5 = 5, 6+7+8+9+10 = 40, 11 = 11. [2002 Q11 · biggest_piece]

5. C (the piece with the 8) — The sums are: 4 = 4, 5 = 5, 6 = 6, 7+8+1+2 = 18, 3 = 3. [2002 Q11 · biggest_piece]

6. B (3 apples) — Take 2 pumpkins and 3 apples off both plates. Left: 1 pumpkin = 3 apples. [2002 Q9 · both_sides]

7. D (Fabienne was born in June) — The only possible matching: Chloe – March; Manon – October; Fabienne – June; Sofia – July. [2002 Q18 · which_true]

8. D — All numbers add up to 45, so the pieces must be 15 + 15 + 15 = 45. Only this picture gives 15, 15, 15. [2002 Q11 · which_picture]

9. B — Only this piece is found in the line. The others don't fit — A: it has a pointed roof where the line has a round dome; C: it has a pointed roof where the line has two small bumps; D: it is a mirror image of a part of the line; E: it is a mirror image of a part of the line. [2002 Q7 · can_belong]

10. B (4) — One bowling ball = 2 footballs = 2 × 2 = 4 tennis balls. [2002 Q9 · chain]

11. B (72 cm) — Side = 36 ÷ 4 = 9 cm. There are 2 cuts; each cut is 9 cm long and is counted twice (once for each piece next to it): 36 + 2 × 2 × 9 = 72 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

12. D (Aisha plays the guitar) — The only possible matching: Aisha – guitar; Hana – harp; Emma – drum; Julien – flute. [2002 Q18 · which_true]

13. A (32 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (10 + 6) = 32 cm. [2002 Q6 · corner_cut]

14. C (3 nuts) — Take 1 coconut and 1 nut off both plates. Left: 1 coconut = 3 nuts. [2002 Q9 · both_sides]

15. D (50 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 4 cm. Perimeter = 2 × (11 + 10) + 2 × 4 = 50 cm. [2002 Q6 · notch]

16. B (1) — Each half must have 17. The ways: 6+2+1+8 | 5+2+7+3. [2002 Q11 · count_ways]

17. C (3) — Pieces 2, 4, 5 fit. Not: 1 (it has the right shapes but in the wrong order); 3 (it has the right shapes but in the wrong order). [2002 Q7 · how_many_fit]

18. A (2) — One ball = 2 marbles, so the left plate is worth 1 × 2 + 1 = 3 marbles. The right plate has 1, so add 3 – 1 = 2. [2002 Q9 · what_to_add]

19. E (Emma plays the violin) — The only possible matching: Emma – piano; Leo – harp; Tom – violin; Fabienne – xylophone. [2002 Q18 · which_not_true]

20. C (3) — Pieces 3, 4, 5 fit. Not: 1 (it steps down where the line steps up); 2 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]